

Thomas Chisica Londoño

Applied Mathematics & Computer Science | Bogotá, Colombia

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RESEARCH PROFILE

Applied mathematics and scientific software for reliable learning systems. Experience in reinforcement learning, recurrent-network experiments and LLM retrieval systems, with strong baselines and held-out evaluation. Interests: reasoning agents, optimisation, formal verification and computational neuroscience.

EDUCATION

Universidad del Rosario, Bogotá

2024–present

BSc in Applied Mathematics and Computer Science. Expected graduation: June 2028.

GPA: 4.11/5.00 | Full-tuition merit scholarship.

Relevant coursework: Machine Learning, Optimisation, Probability & Statistics, Real Analysis, Linear Algebra, Graph Theory and Differential Equations; Numerical Analysis (in progress).

TECHNICAL SKILLS

Programming & ML: Python, PyTorch, Gymnasium; C, R, SQL.

Scientific computing: NumPy/SciPy, SimPy, NetworkX, statsmodels.

LLM systems: retrieval-augmented generation, embeddings, FAISS, FastAPI.

Software & data: Git, Linux, pytest/CI, SQLite, PostgreSQL, Polars, $\text{\LaTeX}$.

SELECTED RESEARCH & TECHNICAL EXPERIENCE

Reinforcement learning for supply-chain control

2025–present

Undergraduate research with Prof. Alexander Garrido, Universidad del Rosario

- Built a thirteen-operation SimPy supply-chain simulation and Gymnasium environments for PPO; added automated tests, source-model checks and contributor documentation.
- Benchmarked 10 PPO checkpoints on 60 held-out streams against a calibration-selected, action-matched static policy; identified comparator sensitivity behind the apparent RL advantage.

 [SCRES: simulation and evaluation code](#)

Recurrent neural networks: connectivity and motor learning

July 2026

Neuromatch Academy, Computational Neuroscience: team project and independent follow-up

- Ran Python/NumPy experiments on 200-unit recurrent networks across four connectivity densities and eight paired seeds, with learning disabled during held-out evaluation.
- Implemented an equal-plasticity control separating connectivity density from trainable-edge count; the original density trend did not recur under this control.

 [Motor-RNN experiments and equal-plasticity control](#)

Witty: retrieval-augmented support prototype

June–December 2025

Lead in Emerging Technologies, Digital Business Ventures; project with Progresión

- Built a FastAPI retrieval pipeline with embeddings, optional FAISS search and LLM rewriting; implemented heuristic answer/escalate routing, not calibrated confidence.

Spectral and geometric analysis of human collective search

2026

Reanalysis of Andrade-Lotero and Goldstone's SODCL experiment and data

- Built a spatial-graph reanalysis and reconstructed past-only features for 45 dyads; checked source metrics and temporal integrity with six tests.
- Implemented basis-invariant Fiedler-subspace projectors; spectral and coordinate energies were nearly redundant in this sample. Predictive advantage remains unestablished.

 [Spectral/SODCL: reanalysis and temporal audit](#)

ADDITIONAL RESEARCH EXPERIENCE

Computational analysis of marmot social networks

2026–present

Research with Dr. Adriana Maldonado-Chaparro, Universidad del Rosario

- Built observation-aware SQLite dyadic tables linking interactions, sightings, residence and relatedness; used exploratory count/mixed-effects models and matched-sample comparisons to examine sampling effects.
- Validated a legacy NetLogo/R individual-based model using Python and native-runtime checks; identified implementation defects in homophily rules and documented reproducibility limitations.

Collective behaviour: experimental software

2025–present

Research with Prof. Edgar Andrade-Lotero, Universidad del Rosario

- Contributed workflow documentation and smoke-test commands to the PsyNet-based Coordinator and Foragers experiment, supporting installation and reproducible execution of the team's code.

SELECTED SOFTWARE PROJECTS

Numerical analysis and scientific computing

2026–present

Coursework implementations and tests

Implemented root-finding and quadrature methods in Python, including bisection, Newton, trapezoidal and Simpson rules; added tests against analytical solutions, error tolerances and convergence behaviour.

HeliOS: educational RISC-V kernel

2025–2026

Contributed to a collaborative C kernel with task scheduling, timer interrupts, a UART shell, synchronisation primitives and heap allocation; includes QEMU/OpenSBI smoke tests.

 [Kernel, architecture and test scripts](#)

ContratIA Abierta: public-procurement review tools

2026

Developed a data-processing and review pipeline for Colombian SECOP open data using Polars, PostgreSQL, MongoDB, FastAPI and Dash, with interpretable prioritisation signals and traceable evidence for human review.

 [Data pipeline and review dashboard](#)

ADDITIONAL TRAINING

Neuromatch Academy, Computational Neuroscience — 68-hour interactive course (2026): neural encoding/decoding, GLMs, dimensionality reduction, neuronal dynamics, Bayesian inference, hidden-state models, reinforcement learning and network causality.

TEACHING, AWARDS & FUNDING

Calculus I Teaching Assistant, Universidad del Rosario

July 2024–July 2025

Led problem-solving sessions and supported first-year students.

[Andrés Bello National Distinction](#)

2024

Colombian Ministry of Education

Ranked 33rd nationally in the official distinction ranking for the 2023 Saber 11 examination (Colombia's national school-leaving examination); one of 50 national recipients. Highest-ranked recipient from Tunja, which ranked first among certified education authorities in the reported 2023 results.

LANGUAGES

Spanish (native); English (IELTS overall band 7.0, 2023).